algorithms 2 homework 4

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May 16, 2023

note
this document and others can be found on my website
`https://mahmoodsheikh36.github.io/`

Question 1. let $G = (V, E, W)$ be an undirected connected and positively weighted graph and let T_1, T_2 be two minimum spanning trees of G. prove that the sum of squares of the weights of T_1 is equal to that of T_2

Answer 1.0.1. at first i thought it was obviously correct, because $x^2 = x^2$ where x is the sum of weights in every mst of G, but then realized that just because the sums of weights are doesnt mean the weights themselves are equal (e.g. $3^2 + 4^2 \neq 5^2 + 2^2$, but according to the lemma we proved in class they are indeed equal, so their summed squares must also be equal

Question 2. let $G = (V, E, W)$ be an undirected connected and positively weighted graph, and given that the weights of the edges of G differ from one another. prove that for every G there is a single mst

Answer 2.0.1. for every two mst's T_1, T_2 in G we know $\#_a(T_1) = \#_a(T_2)$ for every weight a, and since the weights of the edges in the graph are all distinct each weight corresponds to one edge, and so 2 equal sets of weights would refer to the same set of edges, and so $T_1 = T_2$

Question 3. given an undirected positively weighted graph $G = (V, E, w)$, and given an edge $e \in E$

propose an efficient algorithm that returns true if for G there is an mst that contains e and false otherwise. prove the correctness of the algorithm
note: the correctness proof is the essense of the question

Answer 3.0.1. we modify kruskal's algorithm so that it assumes that the mst already contains the edge and build from there, if we succeed in building a full mst with $|V| - 1$ edges then such an mst exists, otherwise it doesnt

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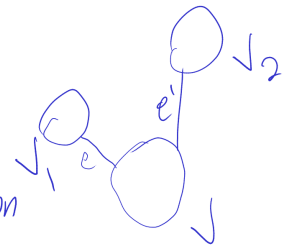
KRAKSUL(G,edge):
    E_out ← {edge}
    UNION/FIND-INIT(n)
    SortE ← Sort(E − {edge})
    for k ← 1 to |E| − 1 do
        (u, v) ← SortE[k]
        fu ← Find(u)
        fv ← Find(v)
        if fu ≠ fv then
            UNION(fu, fv)
            E_out ← E_out + (u, v)
            if |E_out| = |V| − 1 then
                return E_out
    return Null // the mst doesnt exist
```

Question 4. given an undirected positively weighted graph $G = (V, E, w)$
definition: a minimal edge of a vertex v is an edge with the smallest weight of ones that touch v (v is one of their endpoints)
note: a vertex can have multiple minimal edges
prove that every vertex in the graph that has a single minimal edge has to be in every mst of the graph

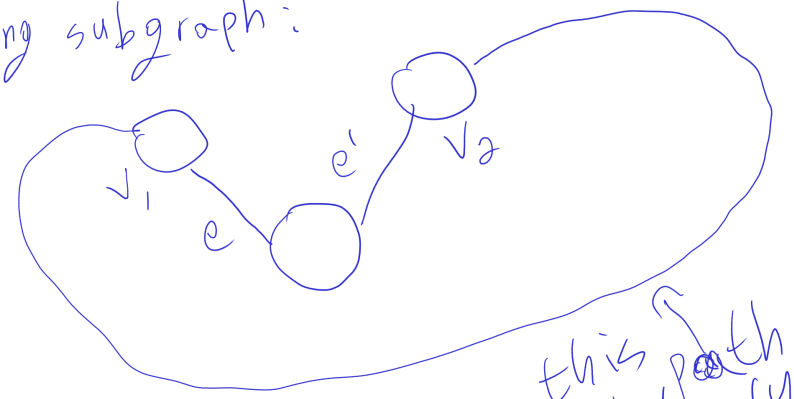
consider the following subgragh of G :-

$w(e) < w(e')$



assume in contradiction
$e \notin T$

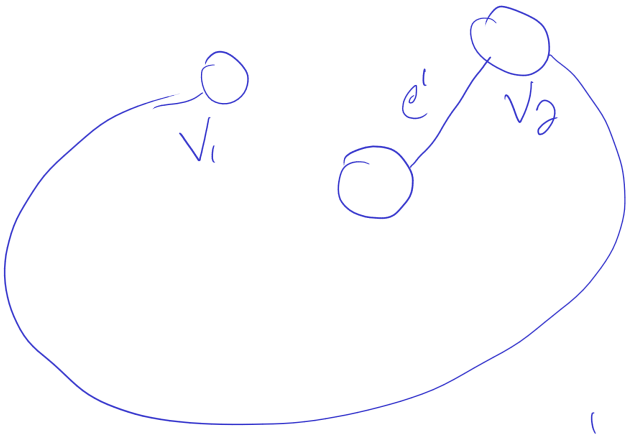
for there to be a minimum spanning tree T there has to be an edge e' in the tree T that connects it to v, because a spanning tree contains all vertices
let v_1 be the vertex on the other endpoint of $v - v_1$
there also has to be a path in T between v_1 and v_2, we add that path to get the following subgraph:



this path is arbitrary

Proof.

so a subtree of T now looks like this:



if were to remove e' and replace it with e we would get the respective subtree of T but with a smaller total weight since $w(e) < w(e')$:

this contradicts with the assumption that T is an MST

*this pattern repeats for every v

□

Question 5. given $G = (V, E, w)$ is an undirected weighted graph, let

$e_1, e_2 \in E$ be two edges of the same minimal weight in G (only these 2 have this weight)

Subquestion 5.1. prove that for every mst of G must contain e_1, e_2

Answer 5.1.1. if the edges dont share a vertex then this trivially follows from the proof in the previous question
if the edges did share a vertex v (v is one of their endpoints), we need to show that its not possible for one to be in an mst without the other, and thats its not possible for neither to be in the mst
assume one of the edges in in the mst but the other isnt, let v_1 be the other endpoint of the first edge which is in the mst and v_2 be the other endpoint of the second edge which isnt (so we're considering 3 vertices in total, like in the proof of the previous question), there has to be a path from v to v_1 but no path can be shorter than the single edge between them, so any spanning tree we build without that edge is not gonna be an mst because there is atleast one edge in it that can be replaced with another edge with the minimum weight to get a tree with smaller total height
assume both edges arent in the mst, let v_1 be the other endpoint of the first edge and v_2 the other endpoint of the second, as we stated before, any tree that wont contain these 2 edges will contain paths to v_1 and v_2 that are longer, and wont have a minimum height, so its not possible that both edges arent in the mst
so both edges have to be in every mst

Subquestion 5.2. is this claim also correct for 3 edges?

Answer 5.2.1. no, consider the case where the three edges form a simple cycle, we can throw out one of the edges to get a tree with a smaller height